

● MEDIUM

● PROBLEM-SOLVING AND DATA ANALYSIS

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# Problem-Solving and Data Analysis

11

10 questions · ~15 min

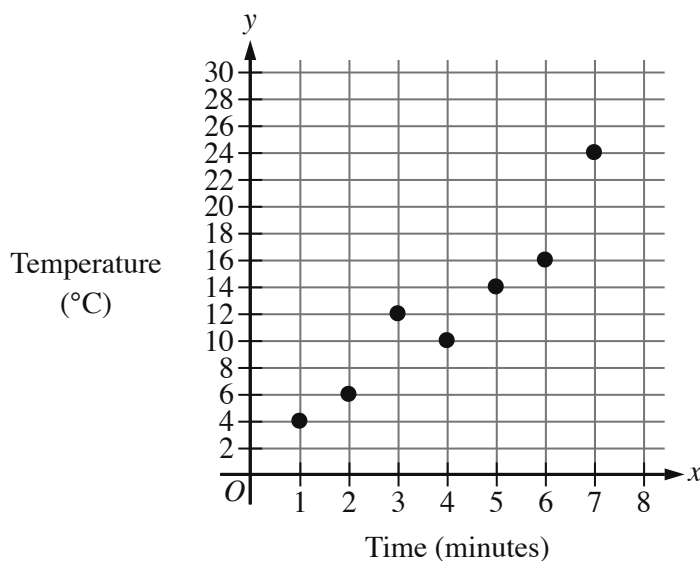
**Q1**

PROBLEM-SOLVING AND DATA ANALYSIS

In a group, 40% of the items are red. Of all the red items in the group, 30% also have stripes. What percentage of the items in the group are red with stripes?

- A. 10%
- B. 12%
- C. 70%
- D. 75%

During a study, the temperature, in degrees Celsius  $^{\circ}\text{C}$ , of the air in a chamber was recorded to the nearest integer at certain times. The scatterplot shows the recorded temperature  $y$ , in  $^{\circ}\text{C}$ , of the air in the chamber  $x$  minutes after the start of the study.



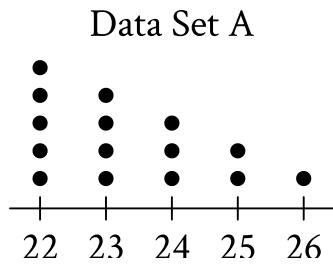
- The scatterplot has 7 data points.
- The data points are in a linear pattern trending up from left to right.
- The data points have the following coordinates:
  - (1 comma 4)
  - (2 comma 6)
  - (3 comma 12)
  - (4 comma 10)
  - (5 comma 14)
  - (6 comma 16)
  - (7 comma 24)

What was the average rate of change, in  $^{\circ}\text{C}$  per minute, of the recorded temperature of the air in the chamber from  $x = 5$  to  $x = 7$ ?

STUDENT-PRODUCED RESPONSE

|   |   |   |   |   |
|---|---|---|---|---|
| . | / | . | / | . |
| 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- The data for the dot plot are as follows:
  - 22: 5 dots
  - 23: 4 dots
  - 24: 3 dots
  - 25: 2 dots
  - 26: 1 dot

Data set A has 15 values and is represented by the dot plot shown. Data set B is created by adding 46 to each of the values in data set A. Which of the following correctly compares the medians and the ranges of data sets A and B?

- A. The median of data set B is greater than the median of data set A, and the range of data set B is equal to the range of data set A.
- B. The median of data set B is greater than the median of data set A, and the range of data set B is greater than the range of data set A.
- C. The median of data set B is equal to the median of data set A, and the range of data set B is greater than the range of data set A.
- D. The median of data set B is equal to the median of data set A, and the range of data set B is equal to the range of data set A.

**Q4**

PROBLEM-SOLVING AND DATA ANALYSIS

For a certain rectangular region, the ratio of its length to its width is 35 to 10. If the width of the rectangular region increases by 7 units, how must the length change to maintain this ratio?

- A. It must decrease by 24.5 units.
- B. It must increase by 24.5 units.
- C. It must decrease by 7 units.
- D. It must increase by 7 units.

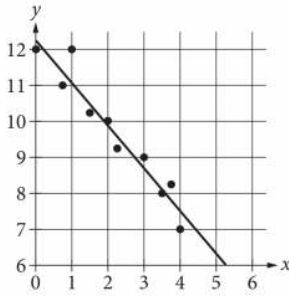
**Q5**

PROBLEM-SOLVING AND DATA ANALYSIS

The population of Greenville increased by 7% from 2015 to 2016. If the 2016 population is  $k$  times the 2015 population, what is the value of  $k$ ?

- A. 0.07
- B. 0.7
- C. 1.07
- D. 1.7

The scatterplot shows the relationship between two variables,  $x$  and  $y$ . A line of best fit for the data is also shown. Which of the following is closest to the difference between the  $y$ -coordinate of the data point with  $x = 1$  and the  $y$ -value predicted by the line of best fit at  $x = 1$ ?



- A. 1
- B. 2
- C. 5
- D. 12

The total mass, in kilograms, of  $r$  identical objects is  $t$ . Which expression represents the total mass, in kilograms, of  $146r$  of these objects?

A.  $146 - t$

B.  $146 + t$

C.  $\frac{t}{146}$

D.  $146t$



Q8

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

13 is  $p\%$  of 25. What is the value of  $p$ ?

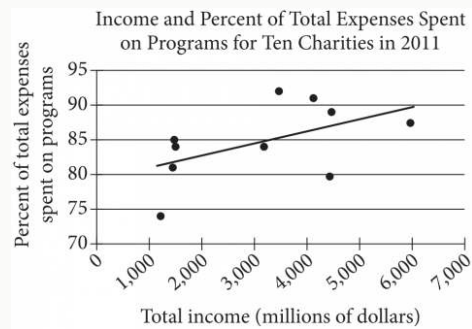
STUDENT-PRODUCED RESPONSE

|  |  |  |  |
|--|--|--|--|
|  |  |  |  |
|--|--|--|--|

$\cdot$   $\frac{\cdot}{\cdot}$   $\frac{\cdot}{\cdot}$   $\cdot$

|   |   |   |   |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

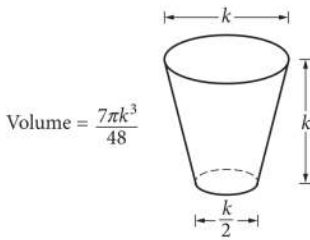


The scatterplot above shows data for ten charities along with the line of best fit. For the charity with the greatest percent of total expenses spent on programs, which of the following is closest to the difference of the actual percent and the percent predicted by the line of best fit?

- A. 10%
- B. 7%
- C. 4%
- D. 1%

Q10

PROBLEM-SOLVING AND DATA ANALYSIS



The glass pictured above can hold a maximum volume of 473 cubic centimeters, which is approximately 16 fluid ounces. Jenny has a pitcher that contains 1 gallon of water. How many times could Jenny completely fill the glass with 1 gallon of water? (1 gallon = 128 fluid ounces)

- A. 16
- B. 8
- C. 4
- D. 3